B GENERALIZATION BOUND FOR ENSEMBLE ADVERSARIAL TRAINING

We provide a formal statement of Theorem 1 in Section 3.4, regarding the generalization guarantees of Ensemble Adversarial Training. For simplicity, we assume that the model is trained solely on adversarial examples computed on the pre-trained models (i.e., we ignore the clean training data and the adversarial examples computed on the model being trained). Our results are easily extended to also consider these data points.

Let $\mathcal{D}$ be the data distribution and $\mathcal{A}_1, \ldots, \mathcal{A}_k, \mathcal{A}^*$ be adversarial distributions where a sample (x, y) is obtained by sampling (x, y_{true}) from $\mathcal{D}$, computing an x^{adv} such that $\|x^{\text{adv}} - x\|_\infty \leq \epsilon$ and returning $(x^{\text{adv}}, y_{\text{true}})$. We assume the model is trained on N data points Z_{train}, where $\frac{N}{k}$ data points are sampled from each distribution $\mathcal{A}_i$, for $1 \leq i \leq k$. We denote $\mathcal{A}_{\text{train}} = \{\mathcal{A}_1, \ldots, \mathcal{A}_k\}$. At test time, the model is evaluated on adversarial examples from $\mathcal{A}^*$.

For a model $h \in \mathcal{H}$ we define the empirical risk

$$\hat{R}(h, \mathcal{A}_{\text{train}}) \coloneqq \frac{1}{N} \sum_{(x^{\text{adv}}, y_{\text{true}}) \in Z_{\text{train}}} L(h(x^{\text{adv}}), y_{\text{true}}) \ , \tag{8}$$

and the risk over the target distribution (or future adversary)

$$R(h, \mathcal{A}^*) \coloneqq \mathop{\mathbb{E}}_{(x^{\text{adv}}, y_{\text{true}}) \sim \mathcal{A}^*} [L(h(x^{\text{adv}}), y_{\text{true}})] \ . \tag{9}$$

We further define the average *discrepancy distance* (Mansour et al., 2009) between distributions $\mathcal{A}_i$ and $\mathcal{A}^*$ with respect to a hypothesis space $\mathcal{H}$ as

$$\text{disc}_{\mathcal{H}}(\mathcal{A}_{\text{train}}, \mathcal{A}^*) \coloneqq \frac{1}{k} \sum_{i=1}^{k} \sup_{h_1, h_2 \in \mathcal{H}} \left| \mathop{\mathbb{E}}_{\mathcal{A}_i} [\mathbb{1}_{\{h_1(x^{\text{adv}}) = h_2(x^{\text{adv}})\}}] - \mathop{\mathbb{E}}_{\mathcal{A}^*} [\mathbb{1}_{\{h_1(x^{\text{adv}}) = h_2(x^{\text{adv}})\}}] \right| \ . \tag{10}$$

This quantity characterizes how "different" the future adversary is from the train-time adversaries. Intuitively, the distance $\text{disc}(\mathcal{A}_{\text{train}}, \mathcal{A}^*)$ is small if the difference in robustness between two models to the target attack $\mathcal{A}^*$ is somewhat similar to the difference in robustness between these two models to the attacks used for training (e.g., if the static black-box attacks $\mathcal{A}_i$ induce much higher error on some model h_1 than on another model h_2, then the same should hold for the target attack $\mathcal{A}^*$). In other words, the *ranking* of the robustness of models $h \in \mathcal{H}$ should be similar for the attacks in $\mathcal{A}_{\text{train}}$ as for $\mathcal{A}^*$.

Finally, let $R_N(\mathcal{H})$ be the average *Rademacher complexity* of the distributions $\mathcal{A}_1, \ldots, \mathcal{A}_k$ (Zhang et al., 2012). Note that $R_N(\mathcal{H}) \to 0$ as $N \to \infty$. The following theorem is a corollary of Zhang et al. (2012, Theorem 5.2):

Theorem 5. *Assume that $\mathcal{H}$ is a function class consisting of bounded functions. Then, with probability at least $1 - \epsilon$,*

$$\sup_{h \in \mathcal{H}} |\hat{R}(h, \mathcal{A}_{\textit{train}}) - R(h, \mathcal{A}^*)| \leq \textit{disc}_{\mathcal{H}}(\mathcal{A}_{\textit{train}}, \mathcal{A}^*) + 2R_N(\mathcal{H}) + O\left(\sqrt{\frac{\ln(1/\epsilon)}{N}}\right) \ . \tag{11}$$

Compared to the standard generalization bound for supervised learning, the generalization bound for Domain Adaptation incorporates the extra term $\text{disc}_{\mathcal{H}}(\mathcal{A}_{\text{train}}, \mathcal{A}^*)$ to capture the divergence between the target and source distributions. In our context, this means that the model h^* learned by Ensemble Adversarial Training has guaranteed generalization bounds with respect to future adversaries that are not "too different" from the ones used during training. Note that $\mathcal{A}^*$ need not restrict itself to perturbation with bounded ℓ_∞ norm for this result to hold.

C EXPERIMENTS ON MNIST

We re-iterate our ImageNet experiments on MNIST. For this simpler task, Madry et al. (2017) show that training on iterative attacks conveys robustness to white-box attacks with bounded ℓ_∞ norm. Our goal is not to attain similarly strong white-box robustness on MNIST, but to show that our observations on limitations of single-step adversarial training, extend to other datasets than ImageNet.

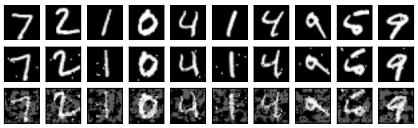

Figure 3: **Adversarial Examples on MNIST.** (top) clean examples. (middle) inputs are rotated by $20°$ and 5 random pixels are flipped. (bottom) The I-FGSM with $\epsilon = 0.3$ is applied.

C.1 A NOTE ON ℓ_∞ ROBUSTNESS ON MNIST

The MNIST dataset is a simple baseline for assessing the potential of a defense, but the obtained results do not always generalize to harder tasks. We suggest that this is because achieving robustness to ℓ_∞ perturbations admits a simple "closed-form" solution, given the near-binary nature of the data. Indeed, for an average MNIST image, over 80% of the pixels are in $\{0, 1\}$ and only 6% are in the range $[0.2, 0.8]$. Thus, for a perturbation with $\epsilon \leq 0.3$, binarized versions of x and x^{adv} can differ in at most 6% of the input dimensions. By binarizing the inputs of a standard CNN trained *without adversarial training*, we obtain a model that enjoys robustness similar to the model trained by Madry et al. (2017). Concretely, for a white-box I-FGSM attack, we get at most 11.4% error.

The existence of such a simple robust representation begs the question of why learning a robust model with adversarial training takes so much effort. Finding techniques to improve the performance of adversarial training, even on simple tasks, could provide useful insights for more complex tasks such as ImageNet, where we do not know of a similarly simple "denoising" procedure.

These positive results on MNIST for the ℓ_∞ norm also leave open the question of defining a general norm for adversarial examples. Let us motivate the need for such a definition: we find that if we first *rotate* an MNIST digit by $20°$, and then use the I-FGSM, our rounding model and the model from Madry et al. (2017) achieve only 65% accuracy (on "clean" rotated inputs, the error is $< 5\%$). If we further randomly "flip" 5 pixels per image, the accuracy of both models drops to under 50%. Thus, we successfully evade the model by slightly extending the threat model (see Figure 3).

Of course, we could augment the training set with such perturbations (see Engstrom et al. (2017)). *An open question is whether we can enumerate all types of "adversarial" perturbations.* In this work, we focus on the ℓ_∞ norm to illustrate our findings on the limitations of single-step adversarial training on ImageNet and MNIST, and to showcase the benefits of our Ensemble Adversarial Training variant. Our approach can easily be extended to consider multiple perturbation metrics. We leave such an evaluation to future work.

C.2 RESULTS

We repeat experiments from Section 4 on MNIST. We use the architectures in Table 5. We train a standard model for 6 epochs, and an adversarial model with the FGSM ($\epsilon = 0.3$) for 12 epochs.

During adversarial training, we avoid the *label leaking* effect described by Kurakin et al. (2017b) by using the model's predicted class $\arg\max h(x)$ instead of the true label y_{true} in the FGSM,

We first analyze the "degenerate" minimum of adversarial training, described in Section 3.3. For each trained model, we compute the *approximation-ratio* of the FGSM for the inner maximization problem in equation (1). That is, we compare the loss produced by the FGSM with the loss of a

Table 5: **Neural network architectures used in this work for the MNIST dataset.** Conv: convolutional layer, FC: fully connected layer.

A	B	C	D
Conv(64, 5, 5) + Relu	Dropout(0.2)	Conv(128, 3, 3) + Tanh	[FC(300) + Relu Dropout(0.5)] $\times 4$
Conv(64, 5, 5) + Relu	Conv(64, 8, 8) + Relu	MaxPool(2,2)	FC + Softmax
Dropout(0.25)	Conv(128, 6, 6) + Relu	Conv(64, 3, 3) + Tanh	
FC(128) + Relu	Conv(128, 5, 5) + Relu	MaxPool(2,2)	
Dropout(0.5)	Dropout(0.5)	FC(128) + Relu	
FC + Softmax	FC + Softmax	FC + Softmax	

Table 6: **Approximation ratio between optimal loss and loss induced by single-step attack on MNIST.** Architecture B' is the same as B without the input dropout layer.

A	A_{adv}	B	B_{adv}	B^*	B^*_{adv}	C	C_{adv}	D	D_{adv}
17%	0%	25%	8%	23%	1%	25%	0%	49%	16%

strong iterative attack. The results appear in Table 6. As we can see, for all model architectures, adversarial training degraded the quality of a linear approximation to the model's loss.

We find that *input dropout* (Srivastava et al., 2014) (i.e., randomly dropping a fraction of input features during training) as used in architecture B limits this unwarranted effect of adversarial training.[3] If we omit the input dropout (we call this architecture B^*) the single-step attack degrades significantly. We discuss this effect in more detail below. For the fully connected architecture D, we find that the learned model is very close to linear and thus also less prone to the degenerate solution to the min-max problem, as we postulated in Section 3.3.

Attacks. Table 7 compares error rates of undefended and adversarially trained models on white-box and black-box attacks, as in Section 4.1. Again, model B presents an anomaly. For all other models, we corroborate our findings on ImageNet for adversarial training: (1) black-box attacks trump white-box single-step attacks; (2) white-box single-step attacks are significantly stronger if prepended by a random step. For model B_{adv}, the opposite holds true. We believe this is because input dropout increases diversity of attack samples similarly to Ensemble Adversarial Training.

Table 7: **White-box and black-box attacks against standard and adversarially trained models.** For each model, the strongest single-step white-box and black box attacks are marked in bold.

	white-box		**black-box**				
	FGSM	R+FGSM	$FGSM_A$	$FGSM_B$	$FGSM_{B^*}$	$FGSM_C$	$FGSM_D$
A	64.7	**69.7**	-	**61.5**	53.2	46.8	41.5
A_{adv}	2.2	**14.8**	6.6	**10.7**	8.8	6.5	8.3
B	85.0	**86.0**	45.7	-	69.9	59.9	**85.9**
B_{adv}	**11.6**	11.1	6.4	**8.9**	8.5	4.9	6.1
B^*	**75.7**	74.1	44.3	**72.8**	-	46.0	62.6
B^*_{adv}	4.3	**40.6**	16.1	14.7	15.0	**17.9**	9.1
C	**81.8**	**81.8**	40.2	55.8	49.5	-	**59.4**
C_{adv}	3.7	**17.1**	9.8	**29.3**	21.5	11.9	21.9
D	92.4	**95.4**	61.3	**74.1**	68.9	65.1	-
D_{adv}	25.5	**47.5**	**32.1**	30.5	29.3	28.2	21.8

While training with input dropout helps avoid the degradation of the single-step attack, it also significantly delays convergence of the model. Indeed, model B_{adv} retains relatively high error on white-box FGSM examples. Adversarial training with input dropout can be seen as comparable to training with a randomized single-step attack, as discussed in Section 4.1.

The positive effect of input dropout is architecture and dataset specific: Adding an input dropout layer to models A, C and D confers only marginal benefit, and is outperformed by Ensemble Adversarial Training, discussed below. Moreover, Mishkin et al. (2017) find that input dropout significantly degrades accuracy on ImageNet. We thus did not incorporate it into our models on ImageNet.

Ensemble Adversarial Training. To evaluate Ensemble Adversarial Training 3.4, we train two models per architecture. The first, denoted $[A\text{-}D]_{adv\text{-}ens}$, uses a single pre-trained model of the same type (i.e., $A_{adv\text{-}ens}$ is trained on perturbations from another model A). The second model, denoted $[A\text{-}D]_{adv\text{-}ens3}$, uses 3 pre-trained models ($\{A, C, D\}$ or $\{B, C, D\}$). We train all models for 12 epochs.

We evaluate our models on black-box attacks crafted on models A,B,C,D (for a fair comparison, we do not use the same pre-trained models for evaluation, but retrain them with different random seeds).

[3] We thank Arjun Bhagoji, Bo Li and Dawn Song for this observation.